ORAL PRESENTATION | TOPIC 7

Synthetic Media & Generative Content

Creating, labelling and detecting AI-generated content responsibly



AI-GENERATED DEMO

ANSA SULTAN

Diploma in Cloud Cyber Security | EduQual Level 3

Presentation Agenda

01

Concepts & Categories

Prompt · model · output

02

How It Works

Diffusion pipeline

03

Architecture

System · data flow · network

04

Workflow & Setup

Process · on-prem · cloud

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Code & Demo

Prompt, settings, output

06

Detect & Disclose

Labels & provenance

07

Risks & Governance

Ethics · legal · policy

08

GitHub, AI Use & Q&A

Project and questions

What Is Synthetic Media?

Synthetic media = content created or altered by AI: text, images, audio and video.



Prompt

Your instruction



Model

Trained AI system



Output

New content

Realistic, fast and cheap to make, so it must be labelled.

Categories & Open-Source Tools

Text

Open LLMs via Hugging Face

Risk: Medium

Image

Stable Diffusion

Risk: High

Voice

Bark · Coqui TTS

Risk: High

Deepfake video

Face-swap · lip-sync

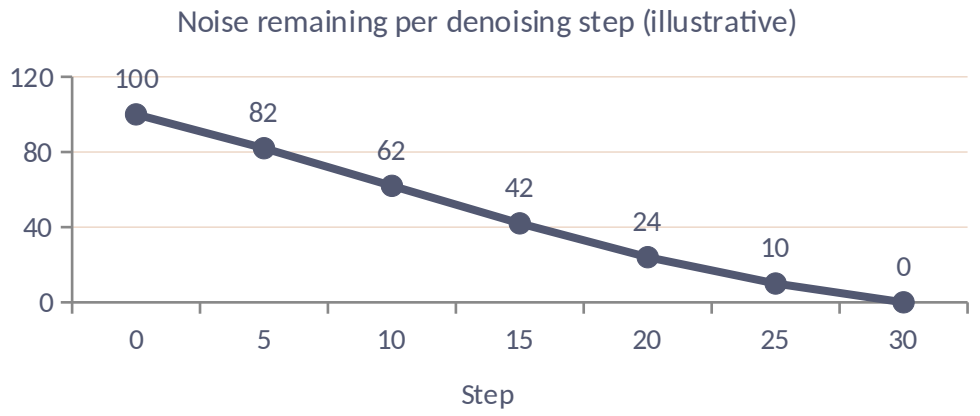
Risk: Highest

Demo focus: image synthesis with Stable Diffusion (open-source, free).

How an Image Model Works

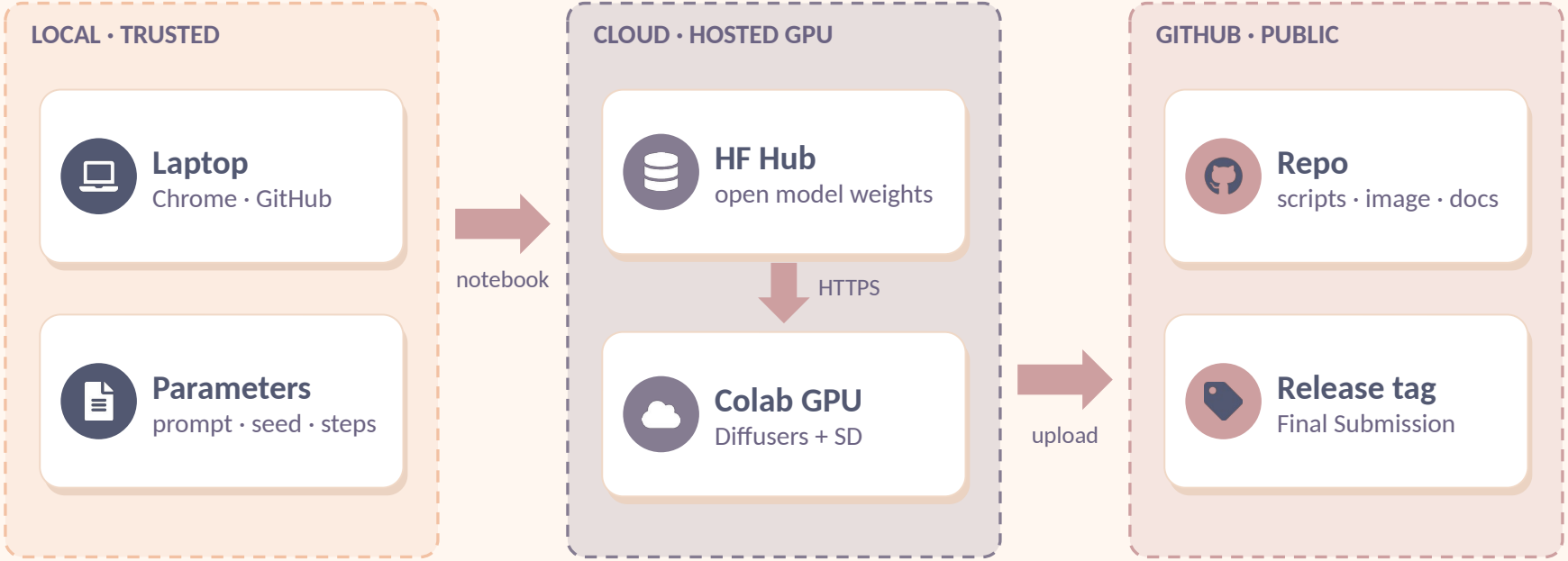
Prompt text → **Text encoder** → **Denoising U-Net** → **VAE decoder** → **Image**

repeats about 30 steps



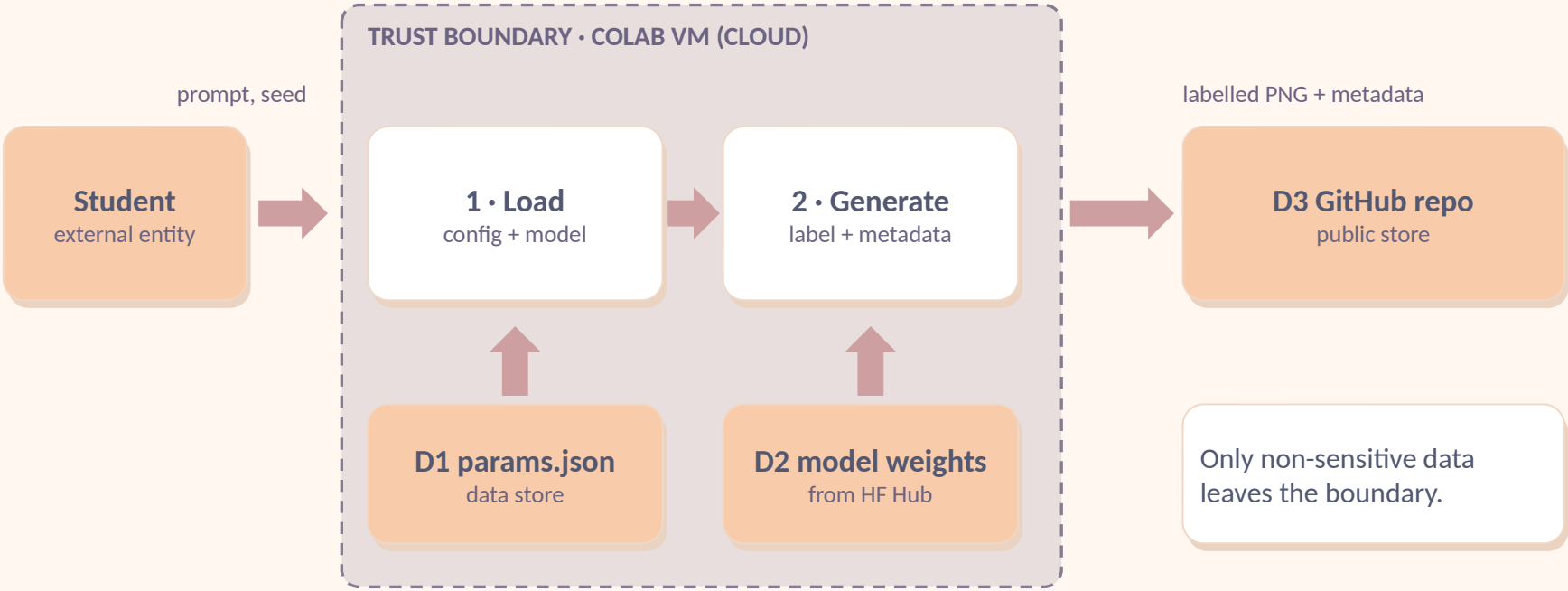
- **Starts from random noise**
- **Prompt guides every step**
- **Noise removed gradually**

System Architecture (Hybrid)

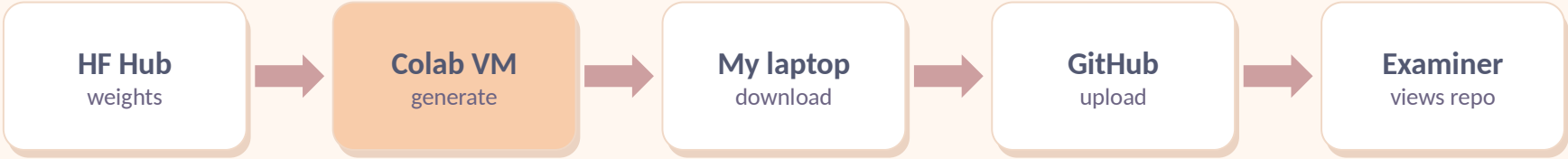


Secrets (API tokens) stay private, never in the repo.

Data Flow Diagram (DFD)

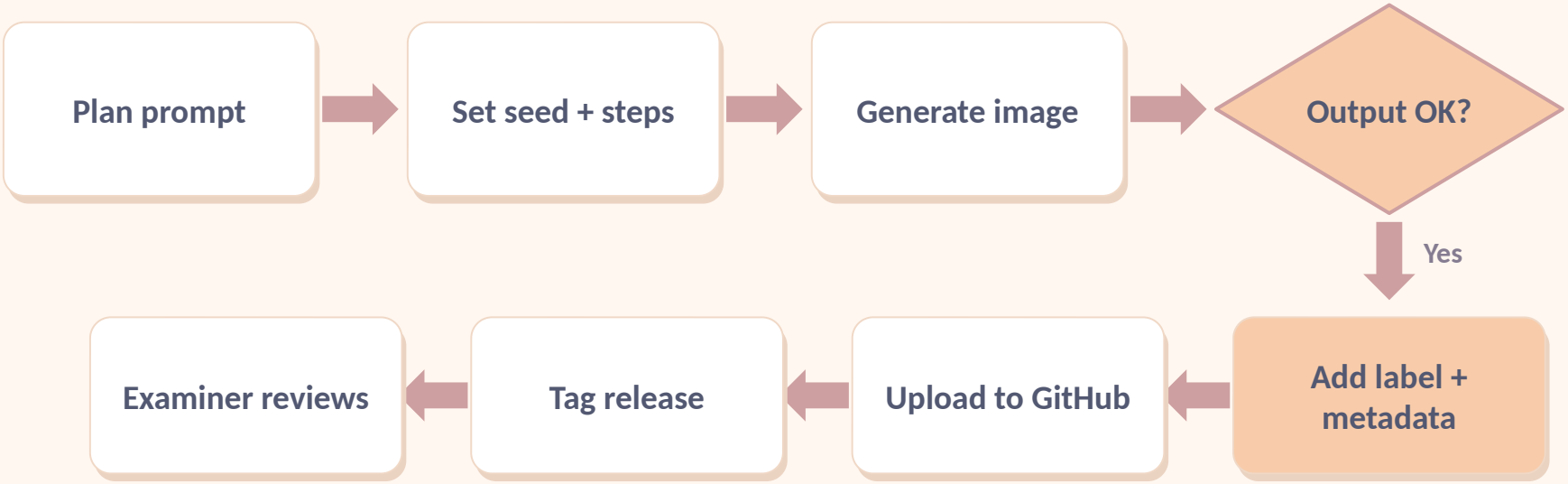


Network & Security Architecture



Layer	Component	Security control
My laptop	Chrome + GitHub account	Strong password + 2-step login
Network	HTTPS / TLS	Traffic encrypted in transit
Cloud compute	Google Colab VM	Temporary, isolated session
Model source	Hugging Face Hub	Open weights, licence checked
Publishing	Public GitHub repo	No secrets; MIT licence

Process Flow: End-to-End Workflow



If not OK: discard it, change the prompt or seed, and generate again.

Deployment: On-Prem, Cloud & Hybrid

	On-Prem (own GPU)	Cloud (Colab)	Hybrid (my setup)
Runs on	Own computer + GPU	Hosted GPU in cloud	Edit locally, run in cloud, publish on GitHub
Needs	Hardware and drivers	Browser only	Browser only
Privacy	Data stays local	Data leaves device	Only non-sensitive data goes online
Scaling	Limited by own GPU	Rent a bigger GPU	Move heavy jobs to cloud



Why hybrid: my laptop has no GPU; the cloud gives one, GitHub shares the result.

Implementation: Generation Script

```
cfg = json.load(open("config/params.json"))
pipe = StableDiffusionPipeline.from_pretrained(
    cfg["model"], torch_dtype=dtype).to(device)
gen = torch.Generator(device).manual_seed(cfg["seed"])
image = pipe(cfg["prompt"],
    negative_prompt=cfg["negative_prompt"],
    num_inference_steps=cfg["steps"],
    guidance_scale=cfg["guidance_scale"],
    generator=gen).images[0]
# label + metadata, then save
image.save("outputs/demo_01.png", pnginfo=meta)
```

Settings live in config/params.json, not in the code.

1. **Install libraries**
2. **Load open model**
3. **Read prompt & seed**
4. **Generate, label, save**

Demo: AI-Generated Output



demo_01.png · AI-generated with Stable Diffusion v1.5

Parameter	Value
Tool	Stable Diffusion v1.5 (open-source)
Prompt	"A watercolor lighthouse at sunrise"
Negative prompt	blurry, low quality, text, watermark
Steps	30
Guidance scale	7.5
Seed	42
Label	Visible + metadata

Same seed + same settings = same image.

Detection, Provenance & Disclosure

Visible label — Demo
"AI-generated" on the image

Embedded metadata — Demo
Model, prompt, seed in file

Provenance standards — Extra
C2PA credentials · watermarks

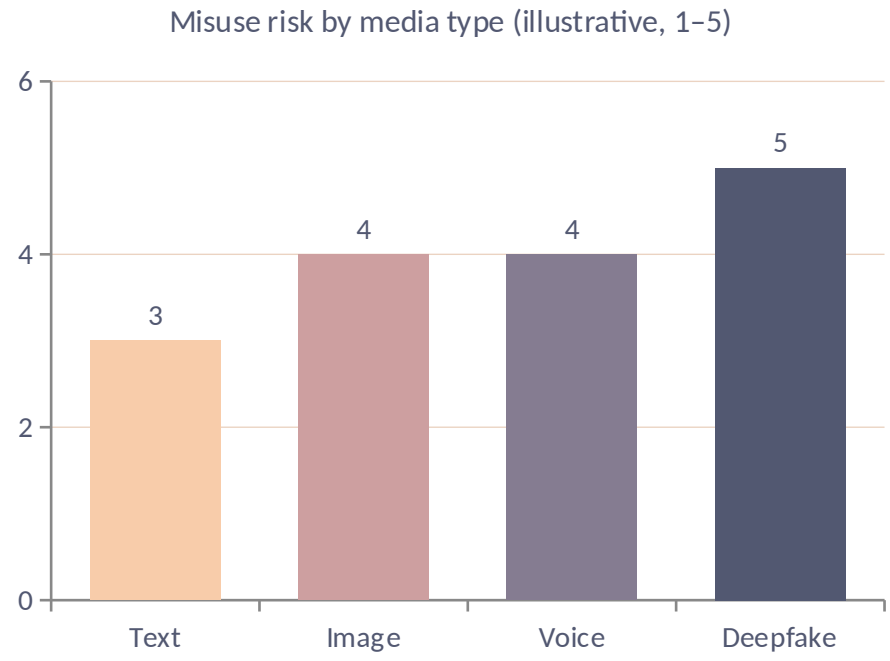
1 · Disclose (write)

```
meta = PngImagePlugin.PngInfo()
meta.add_text("ai_generated","true")
meta.add_text("model","Stable Diffusion v1.5")
meta.add_text("prompt", cfg["prompt"])
meta.add_text("seed", str(cfg["seed"]))
```

2 · Detect (detect_label.py)

```
img = Image.open("demo_01.png")
print(img.text)
```

Ethical, Legal & Misinformation Risks



Misinformation
Fake news, quotes and events

Legal
Copyright, licences, privacy, likeness

Ethical
Consent, bias, loss of trust

Policies, Procedures & Governance

Policies

- Label all AI output
- Use open-licence models
- Consent for real faces and voices
- No sensitive data in prompts

Procedure

1. Plan the prompt
2. Generate with a seed
3. Review the output
4. Label and add metadata
5. Upload and tag release

Standards & Controls

- Check model licence
- Follow copyright and privacy law
- Prompt log in params.json
- DISCLOSURE.md statement

Failure, Recovery, Scalability & Security

1

GPU out of memory

RECOVERY

Lower size, use fp16

2

Model download fails

RECOVERY

Retry or use cached weights

3

Misleading output

RECOVERY

Discard, do not publish

4

Token leaked

RECOVERY

Revoke and rotate key

Scalability
Batch prompts; use bigger GPU

Security
Private tokens; no sensitive data

GitHub Project Submission

```
synthetic-media-and-generative-content/
├─ config/params.json
├─ generate_image.py
├─ detect_label.py
├─ demo_01.png
├─ DISCLOSURE.md
├─ requirements.txt
├─ README.md
├─ LICENSE (MIT)
└─ .gitignore
```

README
purpose · setup · tools · model · prompt · disclosure

13	Public	1
commits	repository	release

- **Setup** Initial commit: README, MIT licence, .gitignore
- **Code** Add script, params and requirements
- **Evidence** Add output image, detect_label.py, DISCLOSURE.md
- **Release** Tag: Final Submission

Responsible & Transparent Use of AI



AI helped me with

- Code samples and scripts
- Configuration examples
- Documentation drafts



I did myself

- Understanding the concepts
- Design and prompt choices
- Explaining and defending

AI is my tool, not a substitute for my understanding.

Q&A Readiness

Architecture decisions

Why hybrid? My laptop has no GPU; Colab has one.

Failure scenarios

Out of memory: smaller size, fp16.

Security implications

Tokens stay private; no secrets in the public repo.

Performance trade-offs

More steps = more detail, but slower.

Real-world deployment

GPU cost, licences, misuse, disclosure rules.

Conclusion & Questions

Prompt → Model → Output

Open tools make it easy

Label + metadata = trust

Consent, licences, verify



Thank you

Questions welcome

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